



Linux For DevOps

1. `id` - This is used to find out user and group names and numeric ID's (UID or group ID) of the current user or any other user in the server.

Example: `id -u root`

2. `cd` - Change Directory: Navigate to a different directory.

Example: `cd /home/user/documents`

3. `pwd` - Print Working Directory: Display the current directory's full path. Example: `pwd`

4. **mkdir** - Make Directory: Create a new directory.

Example: `mkdir new_folder`

5. **rm** - Remove: Delete files or directories.

Example: `rm file.txt`

6. **cp** - Copy: Copy files or directories.

Example: `cp file.txt /backup`

7. **mv** - Move: Move files or directories.

Example: `mv file.txt /new_location`

8. **touch** - Create Empty File: Create a new empty file.

Example: `touch new_file.txt`

9. **cat** - Concatenate and Display: View the content of a file.

Example: `cat file.txt`

10. **nano** - Text Editor: Open a text file for editing.

Example: `nano file.txt`

11. **grep** - Search Text: Search for text patterns in files.

Example: `grep "pattern" file.txt`

12. **find** - Search Files and Directories: Search for files and

directories. Example: `find /path/to/search -name "file_name"`

13. **chmod** - Change File Permissions: Modify file permissions.

Example: `chmod 755 file.sh`

14. **chown** - Change Ownership: Change the owner and group of a file or directory.

Example: `chown user:group file.txt`

15. **ps** - Process Status: Display running processes.

Example: `ps aux`

16. **top** - Monitor System Activity: Monitor system processes in real-time. Example: `top`

17. **kill** - Terminate Processes: Terminate a process using its ID.

Also can use **pkill** to terminate processes based on their name or other attributes.

Example: `kill PID`

`pkill Process_Name`

18. **wget** - Download Files: Download files from the internet.

Example: `wget https://example.com/file.zip`

19. **less** - To view the contents of a file one screen at a time, allowing for easy navigation and search within the file. Example: `less`

`test.log`

20. **tar** - Archive and Extract: Create or extract compressed archive files. Example: `tar -czvf archive.tar.gz folder`

21. **ssh** - Secure Shell: Connect to a remote server securely.

Example: `ssh user@remote_host`

22. **scp** - Securely Copy Files: Copy files between local and remote systems using SSH.

Example: `scp file.txt user@remote_host:/path`

23. **rsync** - Remote Sync: Synchronize files and directories between systems.

Example: `rsync -avz local_folder/ user@remote_host:remote_folder/`

24. **df** - Disk Free Space: Display disk space usage.

Example: `df -h`

25. **du** - Disk Usage: Show the size of files and directories.

Example: `du -sh /path/to/directory`

26. **ifconfig** - Network Configuration: Display or configure network interfaces (deprecated, use **ip**).

Example: `ifconfig`

27. **ip** - IP Configuration: Manage IP addresses and network settings. Example: `ip addr show`

28. **netstat** - Network Statistics: Display network connections and statistics (deprecated, use `ss`).

Example: `netstat -tuln`

29. **systemctl** - System Control: Manage system services using systemd. Example: `systemctl start service_name`

30. **journalctl** - Systemd Journal: View system logs using systemd's journal.

Example: `journalctl -u service_name`

31. **free** - This command displays the total amount of free space available.

Example: `free -m`

32. **at** - Execute Commands Later: Run commands at a specified time. Example: `echo "command" | at 15:30`

33. **ping** - Network Connectivity: Check network connectivity to a host. Example: `ping google.com`

34. **traceroute** - Trace Route: Trace the route packets take to reach a host. Example: `traceroute google.com`

35. - Check Website Connectivity: Check if a website is up.

Example: `curl -Is https://example.com | head -n 1`

36. **dig** - Domain Information Groper: Retrieve DNS information for a domain.

Example: `dig example.com`

37. **hostname** - Display or Set Hostname: Display or change the system's hostname.

Example: `hostname`

38. **who** - Display Users: Display currently logged-in users.

Example: `who`

39. **useradd** - Add User: Create a new user account.

Example: `useradd newuser`

40. **usermod** - Modify User: Modify user account properties.

Example: `usermod -aG groupname username`

41. **passwd** - Change Password: Change user password.

Example: `passwd username`

42. **sudo** - Superuser Do: Execute commands as the superuser.

Example: `sudo command`

43. **lsof** - List Open Files: List open files and processes using them.

Example: `lsof -i :port`

44. **nc** - Netcat: Networking utility to read and write data across network connections.

Example: `echo "Hello" | nc host port`

45. **scp** - Secure Copy Between Hosts: Copy files securely between hosts. Example: `scp file.txt user@remote_host:/path`

46. **sed** - Stream Editor: Text manipulation using regex.

Example: `sed 's/old/new/g' file.txt`

47. **awk** - Text Processing: Pattern scanning and text processing.

Example: `awk '{print $2}' file.txt`

48. **cut** - Text Column Extraction: Extract specific columns from text. Example: `cut -d"," -f2 file.csv`

49. **sort** - Sort Lines: Sort lines of text files.

Example: `sort file.txt`

50. **diff** - File Comparison: Compare two files and show differences.

Example: `diff file1.txt file2.txt`

51. **ls** - List Files and Directories: List the contents of a directory.

Example: `ls -la`

52. **history** - This command is used to view the previously executed command.

Example: `history 10`

53. **cron** - Schedule Tasks: Manage scheduled tasks.

Example: `crontab -e`

54. **ssh-keygen** - This command is used to generate a public/private authentication key pair. This process of authentication allows the user to connect remote server without providing a password.

Example: `ssh-keygen`

55. `nslookup` - This stands for “Name server Lookup”. This is a tool for checking DNS hostname to Ip or Ip to Hostname. This is very helpful while troubleshooting.

Example: `nslookup google.com`

56. `tr` - For translating or deleting characters.

These commands cover a wide range of tasks that are essential for DevOps professionals working with Linux systems. Remember to always refer to the man pages (`man command`) for more detailed information about each command and its options.

Example: `cat crazy.txt | tr "[a-z]" "[A-Z]"`

57. `tnc` - This is “Test Network Connection” command. Mostly used command while troubleshooting. It displays diagnostic information for a connection.

Example: `tnc google.com --port 443`

58. `w` - Displays current user.

59. **su** - Switch User.

Example: `su - root`

60. **ac (All Connections)** — Total connect time for all users or specified users.

Example: `ac john`

61. **tail** — Displays the last part of a file, commonly used to monitor logs in real-time.

Example: `tail monitor.logs`

62. **head** — Displays the first part of a file, often used to quickly see the beginning of a file's content.

Example: `head content.txt`

63. **ip route** — To show or manipulate the IP routing table. Shows clear ip tables rules.

Example: `ip rout`